

MINERAL WELLS, TX, USA

WMO: 722597

Lat: 32.782N

Lon: 98.060W

Elev: 930

StdP: 14.21

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 93985

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.7	24.6	7.6	8.4	32.1	11.4	10.2	37.0	25.8	49.2	23.7	51.2	7.4	350	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	101.9	73.1	99.4	73.3	97.0	73.4	77.5	91.4	76.7	90.6	75.8	89.5	9.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	131.1	82.3	73.1	127.0	81.3	72.2	123.4	80.7	41.6	91.4	40.6	90.7	39.9	89.8	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.9	18.0	14.1	105.1	3.9	3.1	11.2	107.4	8.9	109.2	6.7	111.0	3.9	113.2
			12.7	79.8	3.7	1.9	10.0	81.2	7.8	82.3	5.8	83.4	3.1	84.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.5	45.3	48.8	57.8	65.5	72.7	80.7	83.9	84.1	77.1	66.2	55.8	46.6
	DBStd	15.99	9.70	11.03	9.75	7.96	6.73	4.55	4.11	4.46	5.74	8.54	9.60	9.68
	HDD50	636	208	143	40	5	0	0	0	0	0	5	51	184
	HDD65	2404	612	462	262	87	18	0	0	0	1	90	297	575
	CDD50	6278	62	109	280	470	704	921	1052	1057	813	507	226	77
	CDD65	2573	1	8	37	103	258	471	587	592	364	127	22	3
	CDH74	29915	43	121	415	1034	2526	5483	7458	7470	3796	1300	225	44
	CDH80	15317	4	27	111	339	1015	2867	4284	4307	1830	478	49	6
(14) Wind	WSAvg	8.7	8.6	9.5	10.1	10.5	9.6	8.8	7.8	7.1	6.9	8.1	8.5	8.5
(15) Precipitation	PrecAvg	32.2	1.7	2.1	2.9	3.0	3.9	3.2	2.1	2.3	3.0	3.1	2.6	1.8
	PrecMax	50.7	5.2	8.4	7.4	10.6	9.1	10.8	5.2	6.7	7.2	8.0	9.1	8.7
	PrecMin	17.4	0.1	0.2	0.0	0.3	1.6	0.3	0.0	0.0	0.1	0.2	0.0	0.0
	PrecStd	8.4	1.2	1.9	2.0	2.2	2.1	2.7	1.7	1.5	1.9	2.4	2.3	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.3	83.9	88.0	93.3	96.0	102.2	105.4	105.9	99.3	92.5	85.3	79.3
		MCWB	58.4	61.5	63.6	68.5	72.4	73.0	72.4	73.3	71.7	70.7	65.6	62.0
	2%	DB	73.0	77.5	83.2	87.7	92.5	98.4	102.0	102.1	95.5	89.0	80.0	73.0
		MCWB	56.5	59.2	63.2	66.8	72.3	73.0	72.8	73.4	72.8	69.8	63.9	57.5
	5%	DB	67.6	72.3	78.7	83.7	89.2	95.5	99.1	99.5	92.8	85.4	75.8	68.3
		MCWB	54.6	57.3	61.6	66.3	71.1	73.5	73.6	73.6	72.2	67.8	61.7	55.9
	10%	DB	63.1	67.2	74.2	79.9	85.9	92.9	96.4	97.1	90.1	81.9	71.6	63.2
		MCWB	51.5	55.6	60.2	64.9	70.8	73.5	73.8	73.9	71.7	65.9	60.0	53.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.4	66.4	69.1	73.2	77.3	79.2	78.4	78.2	77.1	74.7	70.0	65.8
		MCDB	70.7	75.6	80.3	83.8	90.4	91.3	92.0	93.8	90.4	85.4	78.6	72.5
	2%	WB	60.9	63.9	66.8	71.2	75.3	77.3	77.4	77.0	75.6	73.0	67.0	62.4
		MCDB	67.7	71.1	76.5	81.4	86.2	89.8	91.5	92.5	88.3	83.5	74.0	69.3
	5%	WB	56.7	61.4	65.1	69.5	74.0	76.2	76.5	76.1	74.7	71.3	64.8	58.6
		MCDB	64.7	68.8	74.3	78.4	84.9	88.6	90.6	91.3	86.5	80.4	71.9	64.7
	10%	WB	52.5	57.7	63.0	68.0	72.7	75.1	75.5	75.2	73.6	69.6	62.6	54.3
		MCDB	61.7	65.5	71.4	76.0	82.9	87.4	89.7	90.1	84.4	76.8	69.8	62.0
(35) Mean Daily Temperature Range	5% DB	MDBR	24.5	23.5	24.2	23.7	21.7	21.7	22.9	23.2	22.6	24.3	24.0	23.3
		MCDBR	32.6	31.6	29.4	28.6	25.8	25.3	27.0	26.8	25.4	27.6	28.2	30.7
		MCWBR	18.7	16.9	13.8	13.0	10.2	7.2	6.7	6.6	8.0	12.0	14.4	17.3
	5% WB	MCDBR	27.0	26.1	22.7	21.5	21.4	21.1	21.9	23.0	20.8	21.9	22.2	23.5
		MCWBR	18.1	16.5	11.8	12.0	10.0	8.2	7.1	6.7	7.5	10.8	14.2	16.3
(40) Clear-Sky Solar Irradiance	taub		0.304	0.314	0.344	0.382	0.416	0.425	0.429	0.421	0.393	0.352	0.327	0.309
	taud		2.519	2.499	2.433	2.348	2.320	2.329	2.361	2.359	2.388	2.487	2.491	2.507
	Ebn at Noon		292	299	297	288	277	273	272	273	277	282	280	283
	Edh at Noon		27	30	35	40	41	41	39	39	36	30	27	26
(44) All-Sky Solar Radiation	RadAvg		954	1135	1491	1802	1925	2152	2172	1977	1659	1306	1007	837
	RadStd		98	174	108	148	193	192	141	128	138	195	130	106

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page